LAB 4

#1

```
#include <iostream>
using namespace std;
class Rectangle
{
private:
    float length;
    float breadth;
public:
    Rectangle()
    {
        length = 0;
        breadth = 0;
    }
    Rectangle(float l, float b)
    {
        length = l;
        breadth = b;
    }
    Rectangle(float x)
    {
        length = x;
        breadth = x;
    }
    float area()
    {
        return length * breadth;
    }
};
int main()
{
    Rectangle r1;
    Rectangle r2(5);
    Rectangle r3(4, 6);
    cout << "Area of r1: " << r1.area() << endl;
    cout << "Area of r2: " << r2.area() << endl;
    cout << "Area of r3: " << r3.area() << endl;
    return 0;
}
```

```
PS C:\Users\lenovo\Desktop\DSA>
Area of r1: 0
Area of r2: 25
Area of r3: 24
```

#2

```
#include <iostream>
using namespace std;

class Rectangle
{
private:
    float length;
    float breadth;

public:
    Rectangle(float l = 0, float b = 0)
    {
        length = l;
        breadth = b;
        cout << "Constructor called\n";
    }

    float area()
    {
        return length * breadth;
    }
    ~Rectangle()
    {
        cout << "Destructor called\n";
    }
};

int main()
{
    Rectangle arr[3] = { Rectangle(),
Rectangle(5), Rectangle(4,6) };

    for(int i = 0; i < 3; i++)
    {
        cout << "Area of object " << i+1 << ": "
<< arr[i].area() << endl;
    }

    return 0;
}
```

```
Constructor called
Constructor called
Constructor called
Area of object 1: 0
Area of object 2: 0
Area of object 3: 24
Destructor called
Destructor called
Destructor called
```

#3

```
#include <iostream>
using namespace std;
class Demo
{
public:
    Demo()
    {
        cout << "Constructor called\n";
    }
```

```
Constructor called
Destructor called
```

```
    ~Demo()
    {
        cout << "Destructor called\n";
    }
};
int main()
{
    Demo obj;
    return 0;
}
```

#4

```
#include <iostream>
using namespace std;

class Sample
{
public:
    Sample()
    {
        cout << "Object Created\n";
    }

    ~Sample()
    {
        cout << "Object Destroyed\n";
    }
};

int main()
{
    int *p = new int;
    *p = 10;
    cout << "Integer value: " << *p << endl;
    delete p;
    float *f = new float;
    *f = 5.5;
    cout << "Float value: " << *f << endl;
    delete f;
    int *arr = new int[3];
    arr[0] = 1;
    arr[1] = 2;
    arr[2] = 3;
    cout << "Integer array: ";
    for(int i = 0; i < 3; i++)
        cout << arr[i] << " ";
    cout << endl;
    delete[] arr;
    // Float array
    float *arr2 = new float[2];
    arr2[0] = 2.5;
    arr2[1] = 4.5;
    cout << "Float array: ";
    for(int i = 0; i < 2; i++)
```

```
PS C:\Users\lenovo\De
Integer value: 10
Float value: 5.5
Integer array: 1 2 3
Float array: 2.5 4.5
Object Created
Object Destroyed
Object Created
Object Created
Object Destroyed
Object Destroyed
```

```
        cout << arr2[i] << " ";
    cout << endl;
    delete[] arr2;
    // Single object
    Sample *obj = new Sample();
    delete obj;
    // Array of objects
    Sample *objArr = new Sample[2];
    delete[] objArr;
    return 0;
}
```